

Projection to 2045: Environmental Performance of MSW Recovery Pathways

Greenhouse gases and fossil energy of six options for 21 Indonesian cities and regencies, 2045 against the existing condition of 2025, with a grid decarbonising linearly to net zero in 2060 — MSW pathway model v3.2

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Key findings

- By 2045 the grid emits 57% less per kWh. This is the only driver that changes emissions per tonne.
- **WtE loses most:** its GHG rises by 177 kg CO₂e/t (median), from 130 to 296, because each exported kWh displaces less coal-based power; its GHG avoided against the landfill falls from 441 to 238 kg CO₂e/t.
- **AD alone** also loses credit (+11 kg/t). **RDF improves** (-20 kg/t) because its electricity use becomes cleaner while the coal it displaces in kilns does not change; **RDF + AD** is almost unchanged (-1 kg/t).
- The **landfill** and **PHB** do not change: their emissions do not involve grid electricity.
- RDF + AD avoids more GHG against the landfill than WtE in 18 of 21 locations in 2045, against 7 of 21 in 2025. Climate benefits of energy recovery that rely on displacing grid power shrink as the grid decarbonises; benefits that rely on keeping organics out of landfills and displacing kiln coal persist.
- Fossil-energy savings follow the same pattern: WtE saves 1.7 GJ/t in 2045 against 4.2 GJ/t in 2025.

1 What is projected and how

The six options are evaluated per tonne of mixed MSW at the facility gate for the existing condition of 2025 and for 2045, for the same 21 locations, with the same model, parameters and composition. Three drivers change between the two years (Table 1); everything else is held constant so that the difference isolates their effect. Both years use the same Monte Carlo random numbers, so changes are computed draw by draw (paired).

$$EF_{grid}(t) = EF_{grid}(2025) \max[0, 1 - \delta(t - 2025)], \quad \delta = \frac{1}{35} \text{ yr}^{-1} \Rightarrow EF_{grid}(2045) = 0.429 EF_{grid}(2025) \quad (1)$$

$$x(t) = x(2025) (1 + e_x)^{t - 2025}, \quad Q(t) = Q(2025) (1 + g)^{t - 2025} \quad (2)$$

Driver	Rule	Factor 2025 to 2045	Affects
Grid emission factor	linear to zero in 2060 ($\delta = 1/35$ per year)	$\times 0.429$	electricity credits of WtE and AD, electricity use of RDF and front-end sorting, fossil CED of grid electricity
Electricity value	real escalation 2%/yr	$\times 1.486$	revenue of WtE and AD
Labour-driven O&M	real escalation 1%/yr	$\times 1.220$	O&M of WtE, RDF and AD plants
Landfill cost	real escalation 2%/yr (third rate, see note)	$\times 1.486$	sanitary landfill and open-dump cost, including residue landfills of every option
Tonnage	each location's growth rate; composition constant	median $\times 1.41$	plant scale (economies of scale) and scale gates

Table 1: Drivers of the projection (data/projection_2045_parameters.csv). CAPEX is constant in real terms. The third escalation rate of the 2/1/2% set is applied to landfill cost, an interpretation to be confirmed. Grid factors in 2045: Jamali 0.877 $\rightarrow$ 0.376, Sumatra 0.832 $\rightarrow$ 0.357, Mahakam 1.128 $\rightarrow$ 0.483 kg CO₂/kWh.

Only the grid driver changes greenhouse gases and fossil energy per tonne: escalation changes prices, not physical flows, and per-tonne emissions do not depend on plant size in this model. Landfill methane, fossil carbon of plastics and coal displaced in cement kilns are unchanged; a cleaner grid makes electricity exported by WtE and AD displace less CO₂, and makes electricity consumed by RDF lines cleaner.

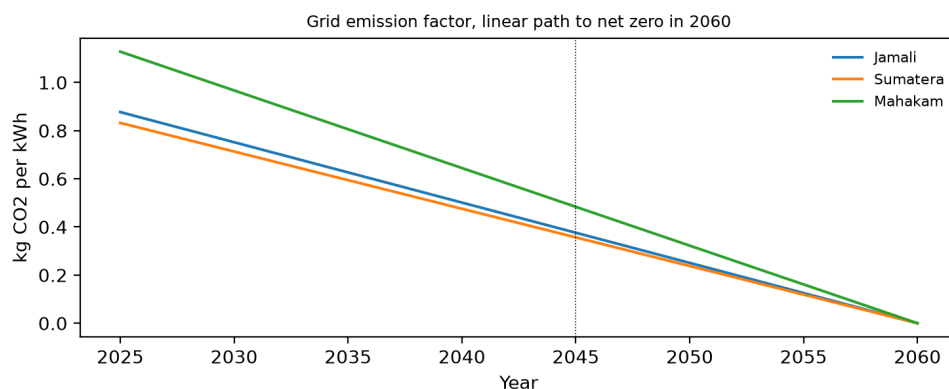


Figure 1: Grid emission factors of the three Indonesian systems on the assumed path (ESDM 2018 values in 2025).

2 Greenhouse gases per tonne, 2025 and 2045

Option	G 2025	G 2045	Change	Change %	Avoided vs OD 2025	Avoided vs OD 2045	Avoided vs SL 2025	Avoided vs SL 2045
SL landfill + flare	557 (424 to 793)	557 (424 to 793)	+0	+0%	425	425	-	-
S1 WtE	130 (-29 to 238)	296 (95 to 454)	+177	+143%	880	675	441	238
S2 RDF	319 (136 to 421)	299 (116 to 401)	-20	-7%	661	681	251	271
S3 AD	389 (192 to 696)	401 (210 to 700)	+11	+2%	582	573	161	150
S4 PHB	543 (414 to 773)	543 (414 to 773)	+0	+0%	439	439	14	14
S5 RDF + AD	153 (23 to 304)	150 (16 to 296)	-1	-1%	810	810	408	409

Table 2: GHG per tonne of MSW (kg CO₂e/t, GWP100), central values, market case: median (minimum to maximum) over 21 locations; avoided = baseline minus option (medians). Change and change % are medians of the per-location changes.

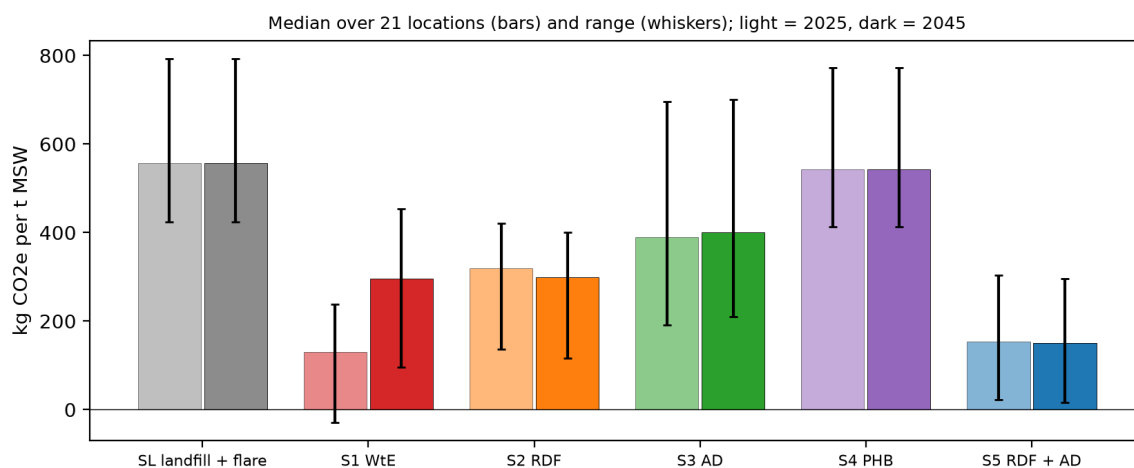


Figure 2: GHG per tonne in 2025 and 2045.

Option	Grid driver	Cost drivers	Tonnage	Total change
SL landfill + flare	+0.0	+0.0	+0.0	+0.0
S1 WtE	+177.5	+0.0	+0.0	+177.5
S2 RDF	-20.0	+0.0	+0.0	-20.0
S3 AD	+10.6	+0.0	+0.0	+10.6
S4 PHB	+0.0	+0.0	+0.0	+0.0
S5 RDF + AD	-1.4	+0.0	+0.0	-1.4

Table 3: Decomposition of the change in GHG (kg CO₂e/t, median over locations), each driver applied alone. Escalation and tonnage do not change emissions per tonne.

Option	Change p5	Change median	Change p95	Locations where GHG rises (median change > 0)
SL landfill + flare	-0	+0	+0	0
S1 WtE	+110	+167	+253	21
S2 RDF	-29	-20	-12	0
S3 AD	-1	+8	+21	21
S4 PHB	-0	+0	+0	0
S5 RDF + AD	-16	-4	+10	4

Table 4: Uncertainty of the change (paired Monte Carlo, 4,000 draws per location): 5th, 50th and 95th percentile of $G(2045) - G(2025)$, median over locations, kg CO₂e/t. The spread comes from the grid multiplier, efficiencies and composition.

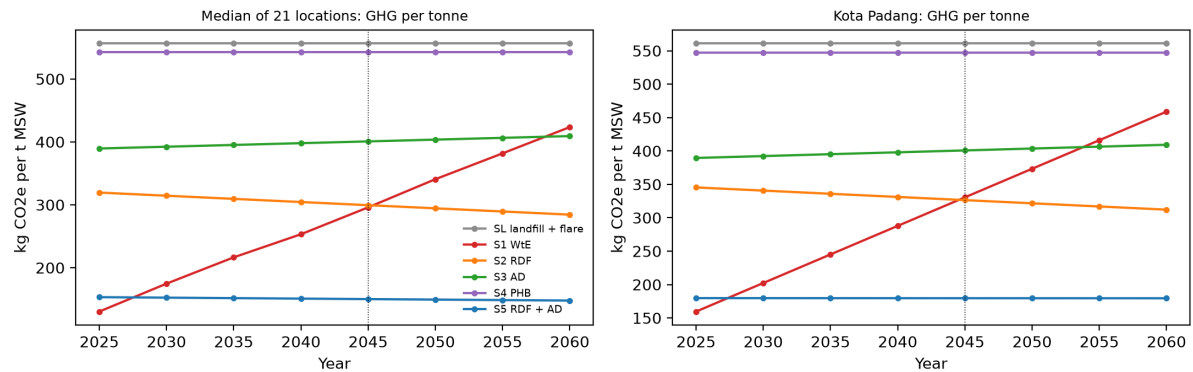


Figure 3: GHG per tonne from 2025 to 2060 on the assumed grid path (central values, all drivers).

3 Fossil cumulative energy demand

Option	CED 2025 (GJ/t)	CED 2045 (GJ/t)	Change (GJ/t)
SL landfill + flare	0.08	0.08	+0.00
S1 WtE	-4.20	-1.70	+2.50
S2 RDF	-3.80	-4.08	-0.28
S3 AD	-0.17	-0.02	+0.15
S4 PHB	-0.57	-0.57	+0.00
S5 RDF + AD	-4.18	-4.23	-0.04

Table 5: Fossil CED per tonne (negative = saving), medians over 21 locations. The fossil primary energy of grid electricity is scaled with the same grid path, so savings from exported electricity shrink.

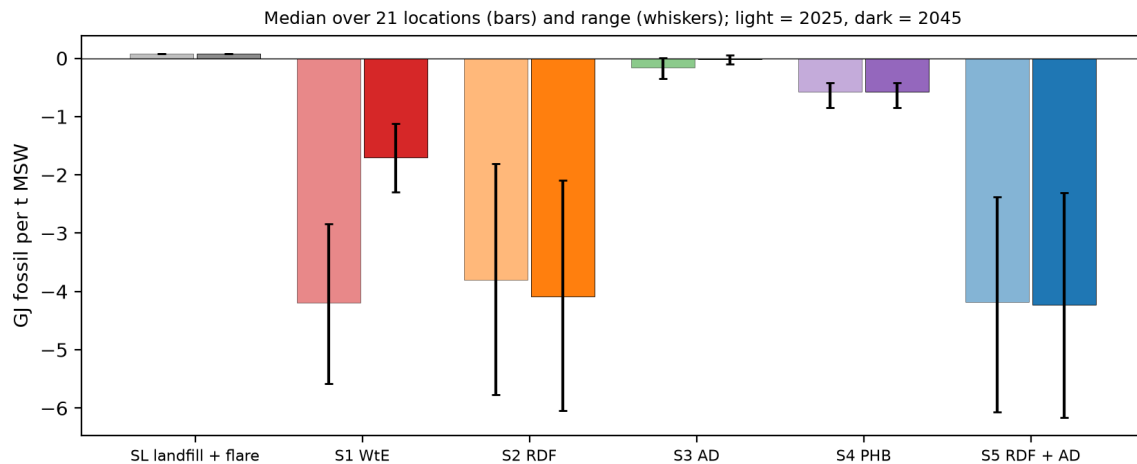


Figure 4: Fossil CED per tonne in 2025 and 2045.

4 Per-location results

Location	S1 2025 / 2045	S2 2025 / 2045	S3 2025 / 2045	S5 2025 / 2045	Avoided vs SL: S5 – S1, 2045
Kota Padang	160 / 331	345 / 326	389 / 401	180 / 180	+151
Kab. Bogor	143 / 290	319 / 299	221 / 238	110 / 115	+175
Kab. Serang	238 / 435	345 / 324	261 / 275	159 / 161	+273
Kota Denpasar	-5 / 163	403 / 383	696 / 700	304 / 296	-133
Kab. Cianjur	92 / 241	345 / 325	377 / 389	176 / 176	+65
Kab. Indramayu	86 / 230	343 / 323	385 / 397	177 / 176	+54
Kota Semarang	130 / 308	296 / 276	384 / 393	153 / 150	+158
Kab. Brebes	237 / 454	254 / 234	335 / 340	141 / 134	+320
Kab. Nganjuk	56 / 207	414 / 394	505 / 515	258 / 256	-49
Kab. Pati	171 / 408	136 / 116	341 / 346	23 / 16	+392
Kab. Magelang	102 / 285	267 / 247	398 / 407	124 / 121	+164
Kab. Pandeglang	116 / 284	289 / 269	192 / 210	68 / 74	+210

Location	S1 2025 / 2045	S2 2025 / 2045	S3 2025 / 2045	S5 2025 / 2045	Avoided vs SL: S5 – S1, 2045
Kab. Pemalang	178 / 376	256 / 236	245 / 259	71 / 73	+303
Kab. Temanggung	135 / 300	307 / 287	372 / 378	182 / 177	+123
Kab. Pekalongan	155 / 364	292 / 272	446 / 455	151 / 148	+216
Kab. Sragen	111 / 296	296 / 276	412 / 422	148 / 145	+151
Kab. Kutai Kartanegara	-10 / 222	361 / 335	544 / 554	222 / 217	+5
Kab. Kendal	-29 / 95	421 / 401	481 / 498	211 / 216	-120
Kab. Wonogiri	13 / 163	342 / 322	440 / 456	141 / 145	+18
Kota Pekalongan	151 / 347	334 / 314	429 / 440	175 / 174	+173
Kota Magelang	142 / 346	272 / 252	457 / 464	145 / 140	+206

Table 6: GHG per tonne (kg CO₂e/t) for the grid-dependent options in each location, and the difference in GHG avoided against the landfill between RDF + AD and WtE in 2045 (positive = RDF + AD avoids more).

5 Interpretation and limits

- Energy recovery evaluated with today's grid factor overstates its long-term climate benefit. A WtE plant commissioned around 2030 operates through the decarbonisation period, so its lifetime-average credit is closer to the 2045 value than to the 2025 value.
- Options whose benefit comes from avoiding landfill methane and displacing kiln coal (RDF, RDF + AD) keep their benefit; this is a robust argument for diversion of organics and RDF where a kiln is in reach.
- The landfill result assumes the same gas collection in both years; better collection or methane regulation would lower it, and kiln decarbonisation (less coal) would reduce the RDF credit. Neither is part of this projection.
- Composition is held at today's proxy; a richer, more plastic-intensive waste in 2045 would raise WtE emissions further.
- Screening-level results; all drivers are user-specified scenario assumptions, not forecasts.